WORKED SOLUTION

STEP III 2020, Question 7

Pure • Second Order Differential Equations

STEP III 2020, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Part (i) shows that a second-order linear equation can sometimes be *factorised* into two first-order linear equations, each of which we can solve with an integrating factor. In part (ii) the work is to find the factorisation: matching coefficients gives a non-linear first-order equation for g, and a power-law guess solves it. After that we just solve two first-order equations in turn, using the initial conditions as we go.

Solution

Part (i)

The second equation gives u in terms of y:

$$u = \frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{g}(x)\, y.$$

Differentiating, using the product rule on $\mathrm{g}(x)y$,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \mathrm{g}'(x)\, y + \mathrm{g}(x)\frac{\mathrm{d}y}{\mathrm{d}x}.$$

Substituting both into $\dfrac{\mathrm{d}u}{\mathrm{d}x} + \mathrm{f}(x)u = \mathrm{h}(x)$:

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \mathrm{g}'(x)\, y + \mathrm{g}(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{f}(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{f}(x)\mathrm{g}(x)\, y = \mathrm{h}(x),$$

and collecting the $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ terms and the y terms gives exactly equation (1).

Part (ii)

An equation for g. Comparing (2) with (1), we need

$$\mathrm{g}(x) + \mathrm{f}(x) = 1 + \frac{4}{x}, \qquad \mathrm{g}'(x) + \mathrm{f}(x)\mathrm{g}(x) = \frac{2}{x} + \frac{2}{x^2}, \qquad \mathrm{h}(x) = 4x + 12.$$

The first gives $\mathrm{f} = 1 + \frac{4}{x} - \mathrm{g}$; putting this into the second eliminates f:

$$\mathrm{g}'(x) + \Big(1 + \frac{4}{x}\Big)\mathrm{g}(x) - (\mathrm{g}(x))^2 = \frac{2}{x} + \frac{2}{x^2}.$$

This is the required first-order equation for g.

A power-law solution. With $\mathrm{g}(x) = kx^n$ the left-hand side is

$$knx^{n-1} + kx^n + 4kx^{n-1} - k^2x^{2n}.$$

Ask yourself How do we choose n before we know k?

Look at the powers of x. The right-hand side contains only x^{-1} and x^{-2}, so the powers x^n, x^{n-1} and x^{2n} must only produce these. The choice $n = -1$ does it: $x^n = x^{-1}$, while x^{n-1} and x^{2n} are both x^{-2}. The left-hand side becomes

$$\frac{k}{x} + \frac{-k + 4k - k^2}{x^2} = \frac{k}{x} + \frac{3k - k^2}{x^2}.$$

Matching the x^{-1} terms gives $k = 2$, and then $3k - k^2 = 6 - 4 = 2$, which matches the x^{-2} term too. So

$$n = -1, \quad k = 2: \qquad \mathrm{g}(x) = \frac{2}{x}, \qquad \mathrm{f}(x) = 1 + \frac{4}{x} - \frac{2}{x} = 1 + \frac{2}{x}.$$

Solving (2). By part (i), (2) is now equivalent to the pair

$$\frac{\mathrm{d}u}{\mathrm{d}x} + \Big(1 + \frac{2}{x}\Big)u = 4x + 12, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2}{x}y = u.$$

Since the conditions are given at $x = 1$, we work on $x > 0$.

The equation for u. The integrating factor is $\exp\int(1 + \frac{2}{x})\mathrm{d}x = \mathrm{e}^{x + 2\ln x} = x^2\mathrm{e}^x$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^2\mathrm{e}^x u) = (4x^3 + 12x^2)\,\mathrm{e}^x.$$

The right-hand side is exactly the derivative of $4x^3\mathrm{e}^x$ (product rule), so

$$x^2\mathrm{e}^x u = 4x^3\mathrm{e}^x + C, \qquad u = 4x + \frac{C\mathrm{e}^{-x}}{x^2}.$$

The value of u at $x = 1$ comes from the initial conditions: $u = y' + \frac{2}{x}y = -3 + 2 \times 5 = 7$. So $4 + C\mathrm{e}^{-1} = 7$, giving $C = 3\mathrm{e}$ and

$$u = 4x + \frac{3\mathrm{e}^{1-x}}{x^2}.$$

The equation for y. The integrating factor is $\mathrm{e}^{2\ln x} = x^2$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^2 y) = x^2 u = 4x^3 + 3\mathrm{e}^{1-x}, \qquad \text{hence} \qquad x^2 y = x^4 - 3\mathrm{e}^{1-x} + D.$$

At $x = 1$: $5 = 1 - 3 + D$, so $D = 7$. Therefore

$$\boxed{y = x^2 + \frac{7 - 3\mathrm{e}^{1-x}}{x^2}}.$$

We used $y(1) = 5$ here and $y'(1) = -3$ in finding C, so both conditions hold.

Checks

- At $x = 1$: $y = 1 + 7 - 3 = 5$. Differentiating, $y' = 2x - 14x^{-3} + 3\mathrm{e}^{1-x}x^{-2} + 6\mathrm{e}^{1-x}x^{-3}$, which at $x = 1$ is $2 - 14 + 3 + 6 = -3$.
- The particular part $y = x^2$ should satisfy (2) on its own: $2 + (1 + \frac{4}{x})2x + (\frac{2}{x} + \frac{2}{x^2})x^2 = 2 + 2x + 8 + 2x + 2 = 4x + 12$. It does.
- Substituting the boxed y into (2) at many points between $x = 0.3$ and $x = 5$ by computer, the two sides agree to within the numerical differentiation error (about 10^{-5}).

Common slips

- Using $y'(1) = -3$ directly as the initial value for u. The initial value for u is $y' + \mathrm{g}y = 7$, not -3.
- Forgetting the $\mathrm{g}'(x)$ term in part (i): g depends on x, so $\mathrm{g}y$ must be differentiated with the product rule.
- Dropping a constant of integration in one of the two stages. There are two constants, which matches the two initial conditions.

Beyond STEP

This is the idea of factorising a differential operator: $(D + \mathrm{f})(D + \mathrm{g})y = \mathrm{h}$, where D means $\mathrm{d}/\mathrm{d}x$. Because g depends on x, the two factors don't commute, and that is why the cross term g' appears. The equation we found for g is a *Riccati equation*. Riccati equations are non-linear, but one known solution is enough to reduce the linear second-order problem to two first-order ones, as here. Boyce and DiPrima, *Elementary Differential Equations*, covers integrating factors, reduction of order and Riccati equations.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.